

WIN-SN-TNH-AOV Environmental Sensor with AOC(4-20mA)-AOV(0-10v) Datasheet

1. Product Overview

The **WIN-SN-TNH-AOC-AOV Sensor Board** is a versatile solution for **measuring Temperature, Humidity, TVOC, and Barometric Pressure** in various settings, including industrial applications and smart homes. It features precise sensors, as detailed in Table 1 and Table 2, and provides analog output options of **10-bit and 12-bit resolution**, ranging from **0 to 10V or 4-20mA**. The board's low power consumption and wide operating temperature range make it suitable for numerous applications, such as **outdoor air quality monitoring, climate control in greenhouses, and environmental monitoring systems**.



Ex: - Product Image

The sensor board offers data output through **analog signals**, making it easy to integrate with various devices and systems. It can be powered by a **DC power supply**, ensuring flexibility and convenience in deployment.

	Sensor Name	Temperature Measuring Range	Humidity Measuring range	Operating Temperature	Measuring Accuracy
☐	AHT21	-40°C ~ 120°C.	0~100% RH.	-40°C ~ 120°C.	0.3°C, 2.0 %RH
☐	AHT20	-40°C ~ 85°C.	0~80% RH.	-40°C ~ 85°C.	0.3°C, 2.0 %RH
☐	SHT21	-40°C ~ 125°C.	0~80% RH.	-40°C ~ 125°C.	0.3°C, 2.0 %RH
☐	SHT25	-40°C ~ 125°C.	0~80% RH.	-40°C ~ 125°C.	0.2°C, 1.8 %RH
☐	BME280	-40°C ~ 85°C.	0~100% RH.	-40°C ~ 85°C.	1.0°C, 3.0 %RH
☐	BME680	-40°C ~ 85°C	0~100% RH.	-40°C to 85°C	1.0°C, 3.0 %RH

Table 1

Sensor Name	Measuring Parameter	Measuring Range	Operating Temperature	Measuring Accuracy
AGS10	TVOC	0 – 9999 ppb	0°C~85°C	(25°C 50%RH) 25% FS
BME280	Barometric Pressure	300 - 1100 hPa	-40°C ~ 85°C.	±1.0 hPa.
BME680	Barometric Pressure	300 - 1100 hPa	-40°C ~ 85°C.	±1.0 hPa.
BME680	IAQ	0-500 IAQ	-40°C ~ 85°C.	±1.0% IAQ
SCD 40	Co2	0-40000ppm	-40°C ~ 85°C.	±5%@2000ppm

Table 2

2. Technical Specifications

- **Measured Parameters:** Temperature and Humidity.
- **Output Type:** Analog Output-10-bit and 12-bit resolution.
- **Output Range:** AOV 0-10V / AOC 4-20mA.
- **Wide temperature range:** -20 °C to +85 °C.
- **Full humidity range:** 0 % to 100 % RH
- **Product dimension:** 85*85*40 mm.
- **Power supply DC:** 12-24V DC @1A (Type –C power USB).

- **Certifications:**



3. Applications

- Outdoor air quality monitoring.
- Climate control in greenhouses.
- Environmental monitoring systems.
- Smart home system.

4. Precautions

- Read all instructions before using the sensor board.
- Keep the sensor board away from water or moisture to prevent damage.
- Do not disassemble or modify the sensor board.
- Use only the specified power sources and mentioned.

5. Power Supply Connection

- **DC Connection**
- Use 12-24V DC @ 1A, Power supply.
- Ensure all connections are tight and secure, **avoid loose connections.**

6. Wire diagram

AO1	Brown- Temperature+
	Black- Temperature-
AO2	Yellow- Humidity+
	White- Humidity-

1. Range for mapping 0-10v or 4-20mA output.

Analog output	Parameter	Range
AO1	Temperature	-40°C ~ 85°C.
AO2	Absolute Humidity	0~130g/m3

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